

Maxwell's thermodynamic relations:

Maxwell's thermodynamic relations are of paramount importance in thermodynamic theorems applicable to simple homogeneous systems.

These relations can be derived by combining the first and the second laws of thermodynamics.

Consider a P-V-T system.

$$TdS = dU + PdV$$

Let us assume that any of the five state functions P, V, T, S, U depend on any two independent thermodynamic variables x and y , so that

$$T(x, y) dS(x, y) = dU(x, y) + P(x, y) dV(x, y)$$

~~Diff~~ Expanding the differentials, we get

$$T \left(\frac{\partial S}{\partial x} \right)_y dx + T \left(\frac{\partial S}{\partial y} \right)_x dy = \left(\frac{\partial U}{\partial x} \right)_y dx + \left(\frac{\partial U}{\partial y} \right)_x dy + P \left(\frac{\partial V}{\partial x} \right)_y dx + P \left(\frac{\partial V}{\partial y} \right)_x dy$$

Now since x and y are independent variables, dx and dy are arbitrary independent variations.

Therefore equating coefficients of dx and dy (2)
~~are~~ we get

$$T \left(\frac{\partial S}{\partial x} \right)_y = \left(\frac{\partial U}{\partial x} \right)_y + P \left(\frac{\partial V}{\partial x} \right)_y \quad \text{--- (1)}$$

$$\text{and } T \left(\frac{\partial S}{\partial y} \right)_x = \left(\frac{\partial U}{\partial y} \right)_x + P \left(\frac{\partial V}{\partial y} \right)_x \quad \text{--- (2)}$$

Differentiating (1) and (2) with respect to y and x respectively, we get

$$\begin{aligned} \left(\frac{\partial T}{\partial y} \right)_x \left(\frac{\partial S}{\partial x} \right)_y + T \cancel{\frac{\partial^2 S}{\partial y \partial x}} - \left(\frac{\partial T}{\partial x} \right)_y \left(\frac{\partial S}{\partial y} \right)_x \\ - T \cancel{\frac{\partial^2 S}{\partial x \partial y}} = \cancel{\frac{\partial^2 U}{\partial y \partial x}} + \left(\frac{\partial P}{\partial y} \right)_x \left(\frac{\partial V}{\partial x} \right)_y \\ + P \cancel{\frac{\partial^2 V}{\partial y \partial x}} \\ - \cancel{\frac{\partial^2 U}{\partial x \partial y}} - \left(\frac{\partial P}{\partial x} \right)_y \left(\frac{\partial V}{\partial y} \right)_x \\ - P \cancel{\frac{\partial^2 V}{\partial x \partial y}} \end{aligned}$$

$$\begin{aligned} \left(\frac{\partial T}{\partial y} \right)_x \left(\frac{\partial S}{\partial x} \right)_y - \left(\frac{\partial T}{\partial x} \right)_y \left(\frac{\partial S}{\partial y} \right)_x \\ = \left(\frac{\partial P}{\partial y} \right)_x \left(\frac{\partial V}{\partial x} \right)_y - \left(\frac{\partial P}{\partial x} \right)_y \left(\frac{\partial V}{\partial y} \right)_x \end{aligned} \quad \text{--- (3)}$$

③

The equation ③ can be symbolically represented by

$$\boxed{\frac{\partial(T, S)}{\partial(x, y)} = \frac{\partial(P, V)}{\partial(x, y)}} \quad \text{--- ④}$$

$$\frac{\partial(T, S)}{\partial(x, y)} = \begin{vmatrix} \left(\frac{\partial T}{\partial x}\right)_y & \left(\frac{\partial S}{\partial x}\right)_y \\ \left(\frac{\partial T}{\partial y}\right)_x & \left(\frac{\partial S}{\partial y}\right)_x \end{vmatrix} \quad \text{is called the Jacobian.}$$

First relation: if $x = T$ and $y = V$, we have

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$$

Second relation: if $x = T$, $y = P$.

$$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$$

Third relation: if $x = S$, $y = V$.

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$$

Fourth relation: if $x = S$, $y = P$.

$$\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$$

Some important properties of the Jacobian:

$$a) \frac{\partial(U, V)}{\partial(x, y)} = \frac{\partial(V, U)}{\partial(y, x)} = - \frac{\partial(V, U)}{\partial(x, y)} = - \frac{\partial(U, V)}{\partial(y, x)}.$$

$$b) \frac{\partial(U, V)}{\partial(x, y)} = \frac{\partial(U, V)}{\partial(p, q)} \frac{\partial(p, q)}{\partial(x, y)}.$$

c) If $\frac{\partial(U, V)}{\partial(x, y)} = 0$, then U and V are functionally related.

[Prove these statements]

Thermodynamic relations with heat capacities: (5)

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Let's consider: $T, V \rightarrow$ independent.

$$S = S(T, V).$$

$$\Rightarrow dS = \left(\frac{\partial S}{\partial T} \right)_V dT + \left(\frac{\partial S}{\partial V} \right)_T dV$$

$$\text{Therefore } \left(\frac{\partial S}{\partial T} \right)_P = \left(\frac{\partial S}{\partial T} \right)_V + \left(\frac{\partial S}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P.$$

$$\Rightarrow T \left(\frac{\partial S}{\partial T} \right)_P - T \left(\frac{\partial S}{\partial T} \right)_V = T \left(\frac{\partial S}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P.$$

$$\left[dQ = T dS \rightarrow \begin{matrix} c_p \\ -c_v \end{matrix} = \left(\frac{\partial Q}{\partial T} \right)_P = \left(\frac{\partial Q}{\partial T} \right)_V \right]$$

$$\Rightarrow c_p - c_v = T \left(\frac{\partial S}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P.$$

$$\Rightarrow \boxed{c_p - c_v = T \left(\frac{\partial P}{\partial T} \right)_V \left(\frac{\partial V}{\partial T} \right)_P}.$$

$$\left(\frac{\partial P}{\partial T} \right)_V = - \left(\frac{\partial P}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P ; \left(\frac{\partial V}{\partial T} \right)_P = - \left(\frac{\partial P}{\partial T} \right)_V \left(\frac{\partial V}{\partial P} \right)_T$$

$$\begin{aligned} \therefore c_p - c_v &= -T \left(\frac{\partial P}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P^2 \\ &= -T \left(\frac{\partial V}{\partial P} \right)_T \left(\frac{\partial P}{\partial V} \right)_T^2. \end{aligned}$$

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Three T-dS equations:

First: $S = S(T, V)$.

$$dS = \left(\frac{\partial S}{\partial T} \right)_V dT + \left(\frac{\partial S}{\partial V} \right)_T dV.$$

$$\begin{aligned} \Rightarrow T dS &= T \left(\frac{\partial S}{\partial T} \right)_V dT + T \left(\frac{\partial S}{\partial V} \right)_T dV. \\ &= C_V dT + T \left(\frac{\partial P}{\partial T} \right)_V dV. \end{aligned}$$

Now, for ideal gas: $PV = nRT$.

$$V \left(\frac{\partial P}{\partial T} \right)_V = nR.$$

$$\Rightarrow T \left(\frac{\partial P}{\partial T} \right)_V = \frac{nRT}{V} = P.$$

$$\Rightarrow \boxed{T dS = C_V dT + P dV.}$$

Second: $S = S(T, P)$.

$$\Rightarrow dS = \left(\frac{\partial S}{\partial T} \right)_P dT + \left(\frac{\partial S}{\partial P} \right)_T dP.$$

$$\Rightarrow T dS = C_P dT + T \left(\frac{\partial S}{\partial P} \right)_T dP.$$

$$= C_P dT - T \left(\frac{\partial V}{\partial T} \right)_P dP.$$

$$= C_P dT - V dP. \text{ (similarly).}$$

The energy equations:

$$U = U(T, V).$$

$$dU = \left(\frac{\partial U}{\partial T} \right)_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV.$$

Now, for PVT system, we know

$$dU = TdS - PdV$$

$$\begin{aligned} \Rightarrow \left(\frac{\partial U}{\partial V} \right)_T &= T \left(\frac{\partial S}{\partial V} \right)_T - P \\ &= T \left(\frac{\partial P}{\partial T} \right)_V - P. \end{aligned}$$

$$\text{Therefore } dU = C_V dT + \left[T \left(\frac{\partial P}{\partial T} \right)_V - P \right] dV$$

Let us consider an ideal gas.

$$P = \frac{nRT}{V} \Rightarrow \left(\frac{\partial P}{\partial T} \right)_V = \frac{nR}{V}.$$

$$\text{Therefore } \left(\frac{\partial U}{\partial V} \right)_T = \frac{nRT}{V} - P = 0.$$

For real gas

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$$\left(p + \frac{a}{v^2}\right)(v-b) = RT.$$

$$p = \frac{RT}{v-b} - \frac{a}{v^2}.$$

$$\left(\frac{\partial p}{\partial T}\right)_v = \frac{R}{v-b}$$

$$\left(\frac{\partial U}{\partial v}\right)_T = \frac{RT}{v-b} - \frac{a}{v^2} p$$

$$= p + \frac{a}{v^2} - p = \frac{a}{v^2}.$$

Second energy equation:

$$U = U(T, p).$$

$$dU = \left(\frac{\partial U}{\partial T}\right)_p dT + \left(\frac{\partial U}{\partial p}\right)_T dp.$$

$$dU = Tds - p dv.$$

$$\left(\frac{\partial U}{\partial p}\right)_T = T \left(\frac{\partial s}{\partial p}\right)_T - p \left(\frac{\partial v}{\partial p}\right)_T$$

$$= -T \left(\frac{\partial v}{\partial T}\right)_p - p \left(\frac{\partial v}{\partial p}\right)_T.$$

$$dU = \left(\frac{\partial U}{\partial T}\right)_p dT - \left[T \left(\frac{\partial v}{\partial T}\right)_p + p \left(\frac{\partial v}{\partial p}\right)_T \right] dp.$$